

Nikhil Advani



+1(716)939-6884 | nikhil.advani.ds@gmail.com | [linkedin.com/in/nikhil-advani/](https://www.linkedin.com/in/nikhil-advani/) | [GitHub](#) | [Portfolio](#)
AWS Certified Data Engineer – Associate | Snowflake SnowPro Core Certified

EXPERIENCE

T-Mobile

Nov. 2025 – Present

AI/ML Engineer

TX, USA

Causal Root-Cause Analysis Engine for Network Operations

- Built an **XGBoost**-based causal regression model on network time-series KPI data to pinpoint root causes of telecom issues
- Applied **SHAP** and **z-score** analysis to rank causal KPIs by severity, replacing manual review of 100+ KPIs with a ranked shortlist
- Partnered with SMEs to select causal KPIs & attach predictions to trouble tickets, cutting identification time from 1 day to 2 hrs
- Engineered **Snowflake Tasks & Stored Procedures** for data pipelines, using **Snowflake Stages** to serve the trained model
- Utilized Snowflake's MCP server with Claude Code to explore schemas & pipelines, cutting onboarding from hours to minutes

Enterprise RAG Pipeline for SharePoint Knowledge Base

- Architected a **RAG** chatbot ingesting 10000+ SharePoint documents via **Airflow**, automating manual search
- Enhanced vector database with semantic chunking, entity extraction, metadata enrichment, improving retrieval quality by **35%**
- Combined vector and keyword search via **Azure AI Search**, powering an AI agent that resolves network queries in under a min

AI Engineer

Jun. 2024 – Oct. 2025

UB Pharmacy & Pharmaceutical Sciences

NY, USA

- Built an end-to-end **RAG** pipeline using LLMs to extract insights from **1000+ clinical trial & drug label documents**
- Engineered **semantic chunking** and **hybrid retrieval (BM25 + MMR)**, boosting query accuracy by 40%
- Optimized latency with **FlashRank reranker**, slashing query time from 60s to 5s
- Deployed the full stack using **AWS EC2** with containerized services for ingestion, storage, and LLM inference via **Bedrock**
- Automated manual research workflows, **reducing turnaround from 36 hours to 5 minutes** with ETL orchestration

Data Scientist

May. 2022 – May. 2023

EPL Limited

Remote

- Developed an XGBoost forecasting model to predict churn & segment high-value users, saving \$1M annually
- Deployed the pipeline on AWS with Docker, DVC, & CI/CD via GitHub Actions, cutting deployment time by 60%
- Monitored production ML system with MLflow & Cloudwatch to track performance, bias, & fairness, reducing downtime
- Optimized SQL data models and ETL workflows in Redshift to improve data accuracy and reduce reporting latency by 40%
- Delivered Power BI dashboards for KPI tracking (churn rate, LTV, retention), adopted by 6 internal teams

EDUCATION

University at Buffalo, Buffalo, NY

Aug. 2023 – Dec. 2024

Masters in Data Science

GPA: 3.90/4.00

- Relevant Coursework: Machine Learning, Statistics, Probability, Data Mining, NLP, Data Visualization, Big Data Analytics

Madhav Institute of Technology and Science, Gwalior

Aug. 2018 – Apr. 2022

Bachelor of Technology, Electrical Engineering

PROJECTS

Expense Tracker MCP Server [\[code\]](#)

- Engineered an MCP server with 6 tools for an AI agent to manage a SQLite expense database via natural language
- Designed structured tool schemas and dynamic SQL queries to keep LLM-generated tool calls reliable on ambiguous inputs
- Built MCP resources (category taxonomy + live date context) to reduce hallucinated categories and misresolved dates

Domain Name Suggester with LLM Fine-Tuning [\[code\]](#)

- Fine-tuned a Llama-3 model with LoRA adapters on a synthetic dataset to generate high-quality business domain names
- Devised an "LLM-as-a-judge" framework with GPT-4 and regex guardrails, filtering inappropriate content with 90%+ accuracy
- Deployed the fine-tuned model as a FastAPI service to provide real-time, production-ready domain name suggestions

Agentic AI-Powered Research Assistant [\[demo\]](#) [\[code\]](#)

- Built a **multi-agent financial chatbot** using Phidata's agentic architecture for automated stock research
- Integrated **Deepseek R1**, **yFinance**, **DuckDuckGo APIs** in a Streamlit dashboard
- Achieved **95% accuracy** in response quality & reduced research time by **40%**

SKILLS

Programming Languages: Python, R, SQL, C++, bash, HTML, CSS, JavaScript

Frameworks & Libraries: Pandas, Numpy, Scikit-learn, Matplotlib, Pytorch, Tensorflow, FastAPI, Langchain, LangGraph, React

Tools & Technologies: Claude Code, MCP, Docker, Git, Apache Spark, Airflow, Jupyter, VS Code, MLOps

Cloud: Amazon Web Services (Redshift, S3, EC2, Bedrock, DMS), Azure (AI Foundry, AI Search, Blob)

Databases: MongoDB, Postgres, Snowflake, SQLite, VectorDBs (ChromaDB, Faiss, Qdrant)